Name _____ Period _____ Date _____

6-2 Lesson Quiz: Exponential Models

short form Savvas numbers 3, 4, 5 (items 1 and 2: full Savvas quiz only); show your work

3. Micah invests $5,280 in an account that earns 4.2% interest, compounded monthly.

Complete the function that models the amount A in the account after t years. Write one tile in each box.

5280	5501	5298.48
$\frac{0.042}{12}$	$\frac{0.042}{6}$	0.042
$12t$	t	$\frac{t}{12}$

$$A(t) = \square\left(1+\square\right)^{\square}$$

Work:

4. Steve invests $1,800 in an account that earns 3.7% annual interest, compounded continuously. What is the approximate value of the account after 10 years?

Ⓐ $2,466 Ⓑ $2,589 Ⓒ $2,601 Ⓓ $2,606

Work:

5. Paul considers investing $8,000 in three different accounts for 6 years.

Option A: 5% simple interest
Option B: 4.5% interest compounded continuously
Option C: 4.75% interest compounded yearly

Word bank:
linear exponential
quadratic constant

After 6 years, Option B, to the nearest dollar, will be worth $_____ more than Option A and $_____ less than Option C.

Option A is a _____ function while Options B and C are _____ functions.

While the _____ function gains value the fastest initially, over time the _____ functions will gain value faster.

Work (use the back if you need more room):

Scoring: Questions 3 and 4, 10 points each; Question 5, bonus up to 5 points. Total out of 20. (Question numbers are the Savvas quiz numbers; items 1 and 2 are on the full Savvas quiz only.)

6-2 Lesson Quiz, short form: answer key

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3. $A(t) = 5280\left(1+\frac{0.042}{12}\right)^{12t}$**.** $P = 5280$, $r = 0.042$, $n = 12$: $A = P\left(1+\frac{r}{n}\right)^{nt}$. Distractor tiles: $5501 \approx 5280(1.042)$ is one year at annual compounding; $5298.48 = 5280\left(1+\frac{0.042}{12}\right)$ is the value after one month; $\frac{0.042}{6}$ and $\frac{t}{12}$ mix up the number of periods. Savvas quiz item 3, DOK 2, Skills Review and Practice F67. Taught by Example 2 (Try It 2 is the same monthly model). **10 points:** 4 for 5280, 3 for $\frac{0.042}{12}$, 3 for $12t$.

4. D, $2,606. $A = Pe^{rt} = 1800e^{0.037(10)} = 1800e^{0.37} \approx 1800(1.447735) \approx \$2{,}605.92$. Choice A is simple interest, $1800(1 + 0.37) = 2466$; B is annual compounding, $1800(1.037)^{10} \approx 2588.57$; C is quarterly compounding, $1800\left(1+\frac{0.037}{4}\right)^{40} \approx 2601.49$. Savvas quiz item 4, DOK 2, F67. Taught by Example 4 (Try It 4). **10 points:** 6 for D, 4 for the work (e raised to the whole product $0.037 \cdot 10$).

5. 80; 89; linear; exponential; linear; exponential. A: $8000(1 + 0.05 \cdot 6) = \$10{,}400$. B: $8000e^{0.045(6)} = 8000e^{0.27} \approx \$10{,}479.72$, so $10{,}480 - 10{,}400 = 80$. C: $8000(1.0475)^6 \approx \$10{,}568.52$, so $10{,}569 - 10{,}480 = 89$. After one year A is $8,400, B $8,368.22, C $8,380: the linear option gains the most at first; the exponential options gain faster over time. Savvas quiz item 5, DOK 2, F67. Taught by Examples 2 and 4 and rehearsed by the packet's Practice (Lena, $5,000, 8 years: a structural parallel with new numbers). **Bonus 5:** 1 for each dollar amount (2); up to 3 for the four words, max(0, 3 − number of wrong words).

Items left off this form (full Savvas quiz only; available for a retake or Form B): Savvas item 1, the town of 4,200 in 1970 decreasing 2.4% per year: $y = 4200(0.976)^t$ and about 0.605% per quarter ($1 - 0.976^{1/4} \approx 0.006055$); DOK 2, F14; taught by Example 1, cut in the fast packet. Savvas item 2, the model $y = a \cdot b^x$ through $(3, 5)$ and $(4, 10)$: answer D, $(5, 20)$ ($b = \frac{10}{5} = 2$, so the next y is $10 \cdot 2 = 20$); DOK 2, F13; taught by Example 5, cut. The quiz has no graphs, so nothing was left off for a figure.

IXL replacement: SmartScore 70 or above on YJW (Compound interest: word problems), 5GU (Continuously compounded interest: word problems) and KLF (Describe linear and exponential growth and decay) replaces this quiz with 20/20.